

Handoff #9: cache guard v2 results

The re-run asked for in handoff #8 is finished: every cache-guard decider, on all three datasets, with the query-only question and the corrected labels. Results, figures 15-16 and the paired tests are committed to main.

TL;DR

- Headline: Jev ties the free similarity threshold where similarity works (LMArena) and beats it where it does not: +27.2 points on GSM-Plus, +4.4 on SearchQueries (both significant).
- Best overall: the threshold -> Jev cascade is best or tied-best on all three datasets (93.3 / 97.4 / 85.7%). On LMArena it asks Jev on only 3% of queries.
- Jev vs frontier: Jev matches GPT-5.6 on SearchQueries (85.6 vs 85.2, tie) and beats it on LMArena (96.3 vs 91.2) at about 1/50 of the price per decision. GPT-5.6 is ahead only on GSM-Plus (+2.6).
- v1 was wrong: handoff #7's "the threshold wins on realistic traffic" came from the old question (deciders graded the stored answer) and the noisy vCache labels. Do not use the handoff #7 cache-guard numbers.
- Cost: \$21.1 on OpenRouter. Balance left is about \$1. Nothing is running.

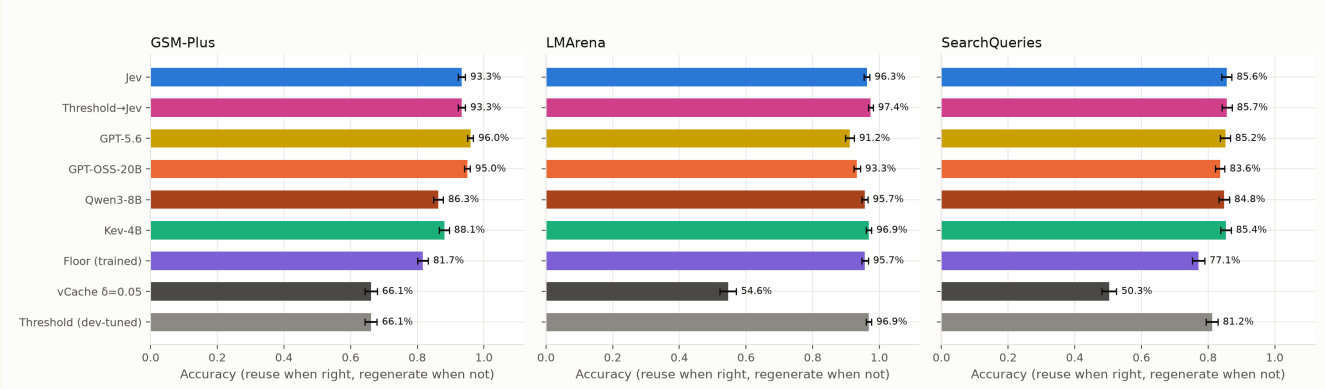
Test accuracy (2,000 queries per dataset; 95% CI in the reports)

decider	GSM-Plus	LMArena	SearchQueries	\$/decision
Jev	93.3	96.3	85.6	~2e-5
Threshold -> Jev cascade	93.3	97.4	85.7	<= 2e-5
GPT-5.6	96.0	91.2	85.2	~1e-3
GPT-OSS-20B	95.0	93.3	83.6	~4e-5
Kev-4B (local)	88.1	96.9	85.4	local
Qwen3-8B (local)	86.3	95.7	84.8	local
Similarity threshold (dev-tuned)	66.1	96.9	81.2	free
Floor (trained on corrected dev)	81.7	95.7	77.1	free
vCache (delta 0.05)	66.1	54.6	50.3	free

Reuse is correct for 34% of GSM-Plus, 93% of LMArena and 58% of SearchQueries test queries, so "always reuse" scores 34 / 93 / 58. vCache is scored on its own labels (it can serve a different cached entry than the one we relabelled).

Figure 16 · accuracy per dataset

Figure 16. Cache-guard accuracy with 95% CI (clustered by class)



Bars are test accuracy; whiskers are 95% CIs clustered by class (a SearchQueries id_set or a GSM-Plus seed question), so near-duplicate queries are not counted as independent.

- GSM-Plus (math variants: a changed number changes the answer): similarity cannot see the difference, so the threshold and vCache never reuse (66.1% = always regenerate). Every model reading the queries does far better.
- LMArena (paraphrased chat prompts, 93% reusable): similarity already works; Jev, Kev and the threshold are within noise of each other. GPT-5.6 is the weakest LLM here because it regenerates too often (8.4% needless regenerations).
- SearchQueries (short web searches): all deciders reading the queries sit at 84-86%; the threshold is 81.2%, and gets there only by serving a wrong answer on 14.4% of queries (Jev: 3.8%).

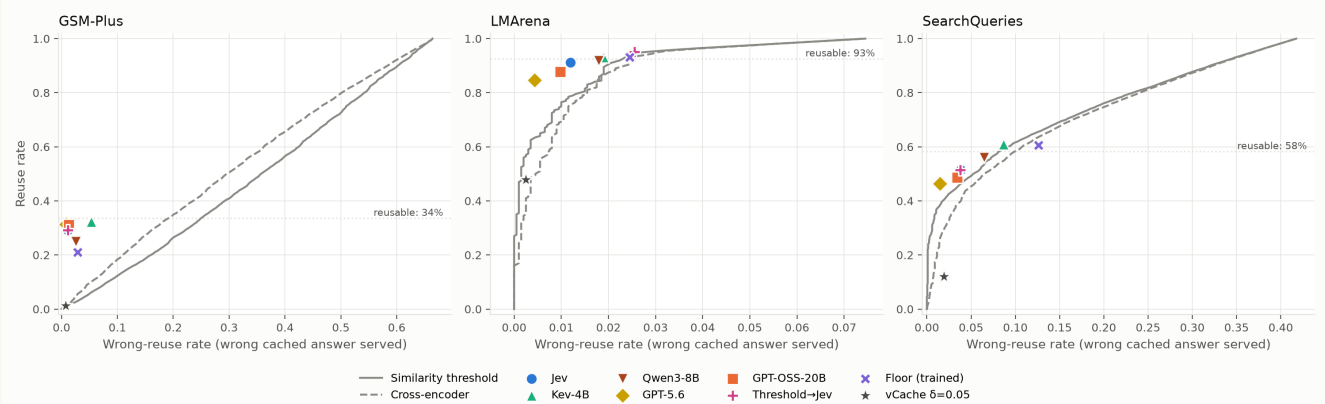
Paired differences (same queries, bootstrap over classes)

dataset	a - b	accuracy diff [95% CI]	wrong-reuse diff [95% CI]
GSM-Plus	Jev - threshold	+27.2* [+25.5, +29.1]	+0.8* [+0.3, +1.4]
GSM-Plus	Jev - floor	+11.7* [+10.1, +13.3]	-1.7* [-2.5, -0.9]
GSM-Plus	Jev - Kev-4B	+5.2* [+4.0, +6.6]	-4.2* [-5.2, -3.2]
GSM-Plus	GPT-5.6 - Jev	+2.6* [+1.9, +3.4]	-0.3 [-0.6, +0.0]
LMArena	threshold - Jev	+0.6 [-0.4, +1.6] (tie)	+1.7* [+1.2, +2.3]
LMArena	cascade - threshold	+0.5* [+0.2, +0.9]	-0.4* [-0.6, -0.1]
LMArena	Kev-4B - Jev	+0.6 [-0.3, +1.4] (tie)	+0.7* [+0.3, +1.1]
SearchQueries	Jev - threshold	+4.4* [+2.3, +6.4]	-10.6* [-12.1, -9.1]
SearchQueries	Jev - floor	+8.5* [+6.5, +10.5]	-8.8* [-10.2, -7.3]
SearchQueries	GPT-5.6 - Jev	-0.4 [-1.5, +0.8] (tie)	-2.3* [-3.1, -1.6]

Points; * = the 95% CI excludes 0. A negative wrong-reuse difference means a serves fewer wrong cached answers. Full table: [results/replay/cacheguard-paired.md](#).

Figure 15 · how much is reused vs. how often it is wrong

Figure 15. Cache guard: reuse vs. wrong reuse (up and to the left is better)



Each point is one decider; the lines are the similarity threshold and the cross-encoder at every possible cut-off. Up and to the left is better: more cache hits, fewer wrong answers served. The dotted line is the share of queries where reuse is correct.

- A point above the line beats every threshold setting, not just the dev-tuned one. On GSM-Plus all query-reading deciders are far above it; on SearchQueries Jev, GPT-5.6 and GPT-OSS sit above it; on LMArena the LLM deciders sit above it at the low-error end.
- vCache (the star) is very cautious: it rarely serves a wrong answer but regenerates almost half the reusable queries on LMArena and SearchQueries, so its accuracy is the lowest.

What changed from v1 (handoff #7) to v2 (this run)

accuracy	GSM-Plus v1 -> v2	LMArena v1 -> v2	SearchQueries v1 -> v2
Jev	93.3 -> 93.3	71.2 -> 96.3	71.3 -> 85.6
GPT-5.6	96.9 -> 96.0	70.6 -> 91.2	71.0 -> 85.2
Threshold (dev-tuned)	66.1 -> 66.1	92.9 -> 96.9	70.4 -> 81.2
Floor	81.7 -> 81.7	90.9 -> 95.7	67.8 -> 77.1

- Question (LMArena): v1 asked whether the stored answer was fully correct, so Jev and GPT-5.6 refused to reuse answers they judged imperfect, even for identical prompts. Asking only "same request?" fixed that: +25 points for Jev.
- Labels (SearchQueries, LMArena): the corrected labels lift every method, the threshold included. Jev's lead over the threshold on SearchQueries is real under the corrected labels (+4.4*), and it was a tie (+0.9) under the old ones.
- GSM-Plus labels did not change and the old question rarely mattered there, so its numbers barely moved.

What happened during the run

- Complete: Jev, Kev-4B, Qwen3-8B 10,000 decisions per dataset (5 repeats x 2,000), GPT-5.6 6,000 (x3), GPT-OSS-20B 5x. Failures are at most 0.4% (GPT-OSS 25-41 per dataset, GPT-5.6 2 on SearchQueries); Jev has none.
- Fix, 284ace2: cacheguard_api.slm now sets per-dataset \$/call estimates. The archived v1 folders left no cost history, so the spending guard would have refused every job.
- Credit: GSM-Plus GPT-5.6 was refused by the guard at a \$7.87 balance; it was re-run in two halves with --limit (the guard's split rule): \$6.67, 0 errors.
- Cluster: Gilbreth maintenance stalled standby for days, so jobs moved to the group's normal QoS (A100-80GB, same hardware as before). The dependent analysis job was cancelled by the scheduler, so its commands were run by hand.
- Jev overload: 13 SearchQueries calls hit HTTP 529 (Jev overloaded). These were infra errors, retried on resume, and all succeeded.
- Git: a session restart replayed the last steps, which left two commits with the same message (7d077cd, 0bc01a3) and duplicate DECISIONS rows, later merged (88dde32). The data is not affected.

Where we are

Step	What	Status
Agent control	arm 0, U4 in-loop A-E, U5 figures 1-14	DONE (handoffs #4-#6)
Router	R1-R6, rigor pass, figure 17	DONE (handoff #7)
Cache guard v2	query-only question, corrected labels, all deciders, paired tests, figures 15-16	DONE (this handoff)
Label spot-check	human check of a sample of the corrected labels	TODO — needs a person
Paper	cache-guard tables, figures 15-16, claims and prose from these reports	TODO

What's next

- Human spot-check: a person (not an LLM) labels about 50 changed + 50 unchanged pairs per dataset (LMArena, SearchQueries) without seeing either label; report agreement with the corrected and with the original labels.
- Paper: replace every cache-guard number from handoff #7 with the v2 reports; state the query-only assumption (handoff #8, page 3) and the label write-up (results/label-corrections/README.md).
- Credit: load OpenRouter before any new paid run. Repeating this whole run costs about \$21.

Where the results are

```
results/replay/cacheguard-<dataset>/report/summary.md # per-dataset tables
results/replay/cacheguard-paired.md # paired differences
results/replay/cacheguard-paper_figures/ # fig15, fig16 (png + pdf), points.json
results/replay/cacheguard-*/v1-answer-question/ # old runs, not used
DECISIONS.md # row dated 2026-10-08
```